

# Younes Essafouri

 [linkedin.com/in/younes-essafouri](https://www.linkedin.com/in/younes-essafouri)  [younes.essafouri@hotmail.com](mailto:younes.essafouri@hotmail.com)  [github.com/YounesEssafouri](https://github.com/YounesEssafouri)

## EDUCATION

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| <b>Institut de Mathématiques de Toulouse (IMT)</b><br><i>PhD in Explainable AI for Weather Forecasting</i>                    | Oct 2025 – Oct 2028 |
| <b>KTH Royal Institute of Technology</b><br><i>Exchange Program in Machine Learning</i>                                       | Aug 2024 - Jan 2025 |
| <b>Grenoble INP-Ensimag, UGA</b><br><i>Engineer's Degree in Computer science and Applied Mathematics</i>                      | Aug 2022 - Aug 2025 |
| <b>Preparatory classes</b><br><i>Two-year program that prepares students for the entrance exams to the top French schools</i> | Sep 2020 - Jul 2022 |

## RELEVANT COURSEWORK

**Mathematics:** Calculus, Linear Algebra, Probability, Statistics, PDEs, Numerical Methods.  
**Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, C++ for Applied Mathematics.  
**Machine Learning:** Artificial Neural Networks, Deep Learning, Statistical Machine Learning, Modern Methods of Statistical Learning, Advanced Machine Learning.

## SKILLS

**Languages:** Python, C/C++, Java, SQL, R,  $\text{\LaTeX}$   
**Tools:** Git/GitHub, Unix Shell, Jupyter, VS Code, R Studio, IntelliJ IDEA, Pycharm, C Lion  
**Libraries:** PyTorch, PyG, TensorFlow, Pandas, NumPy, Matplotlib, Scikit-learn, Seaborn

## EXPERIENCES

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| <b>Research Intern</b><br><i>Dassault Systèmes</i> <ul style="list-style-type: none"><li>Investigated physics-informed machine learning methods for cardiac physics modeling.</li></ul>                                                                                                                                                                                                                                            | Feb 2025 - Aug 2025 |
| <b>AI Researcher</b><br><i>KTH AI Society</i> <ul style="list-style-type: none"><li>Collaborated with researchers from KTH on projects involving Physics-Informed Machine Learning.</li><li>Engaged in discussions on applying machine learning to scientific domains, including fluid dynamics and surrogate modeling.</li></ul>                                                                                                  | Oct 2024 - Jan 2025 |
| <b>Machine Learning Research Intern</b><br><i>Hubert Curien laboratory</i> <ul style="list-style-type: none"><li>Developed physics-informed neural network models to replace computationally expensive numerical solvers for the Kuramoto-Sivashinsky equations.</li><li>Explored continuous domain learning with neural operators, focusing on Fourier Neural Operators (FNOs) for efficient spatial-temporal modeling.</li></ul> | Jun 2024 - Aug 2024 |

## PROJECTS

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| <b>Embedded Topic Modeling</b>   <i>Python, Pytorch</i> <ul style="list-style-type: none"><li>Implemented the Embedded Topic Model (ETM) from scratch, based on the paper “<i>Topic Modeling in Embedding Spaces</i>” by Adji B. Dieng et al. (ICML 2019).</li><li>Performed comparative analysis with Latent Dirichlet Allocation (LDA) to evaluate model performance and interpretability.</li><li>Explored applications of topic models beyond text classification, reproduced results on the 20 Newsgroups dataset.</li></ul> | Nov 2024- Dec 2024 |
| <b>Music Classification using Machine Learning Techniques</b>   <i>Python</i> <ul style="list-style-type: none"><li>Developed a system to classify songs using high-level features from Spotify’s web API, implementing different machine learning models.</li></ul>                                                                                                                                                                                                                                                              | Sep 2024-Oct 2024  |
| <b>Utilizing Neural Networks for Solving Finance-Related PDEs</b>   <i>Python, Tensorflow, Numpy</i> <ul style="list-style-type: none"><li>Employed the Feynman-Kac method to solve the Black-Scholes equation and utilized Python to develop models for pricing Call options numerically. This involved using neural networks to address finance-related partial differential equations.</li></ul>                                                                                                                               | Feb 2024-May 2024  |
| <b>Software Engineering Project</b>   <i>Assembly Language, Java, Maven, ANTLR</i> <ul style="list-style-type: none"><li>Developed a Java-based compiler for Deca, an object-oriented sub-language of Java.</li><li>Developed a Math library to accurately compute trigonometric functions in Deca.</li></ul>                                                                                                                                                                                                                     | Dec 2023-Jan 2024  |

## LANGUAGE PROFICIENCIES

English (Professional), French (Professional), Arabic (Native)